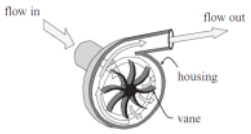
Pump Characteristics Curves

Theory & Background

Pumps → mechanical devices that allow for fluid transportation via pipes (Section 9.2.4.1)

↳ Centrifugal pumps are a common type → transfer rotational energy (from electric motor) to fluid movement

+ Working principle is energy balance! Electric energy → shaft work (by pump to fluid) → affects speed, pressure & other fluid characteristics
+ Pump manufacturers need to characterize a pump → pump operation under various conditions [Pump curves!]



$\Delta E = Q + W$ (W: work done on system by environment)
- ΔE: total energy of system
- Q: Heat from environment to the system via boundary

Energy balance on CV: $\frac{dE_{cv}}{dt} + \int_{cs} (\hat{n}\cdot\underline{v})\rho\hat{E}\,dS = \dot{Q}_{in,cv} - \dot{W}_{by,cv}$ $\quad [E_{cv} = U + E_k + E_p]$

- Closed system
- Open system (mass flow in & out)

↓ Some assumptions

$\Delta E_p + \Delta E_k + \Delta U = Q_{in,cv} - W_{by,cv}$ $\quad (W_f = F\cdot d = (P\cdot A)\cdot d)$

$W_{by,cv}$:
- (W_s) Shaft work (work performed by fluid due to moving parts)
- Flow work (W_f) (work done by fluid as it enters & leaves the system)

$H = U + PV$ ↓

SS Macro Energy balance:

$$\Delta E_p + \Delta E_k + \Delta H = Q_{in,cv} - W_{s,by\,CV}$$

↓ Single in, single out, Incompressible fluid

$$\dot{m}\left(\frac{\langle v\rangle_2^2}{2} - \frac{\langle v\rangle_1^2}{2}\right) + \dot{m}g(z_2 - z_1) + \dot{m}\left(\hat{U}_2 - \hat{U}_1 + \frac{P_2}{\rho} - \frac{P_1}{\rho}\right) = \dot{Q}_{in,cv} - \dot{W}_{s,by\,CV}$$

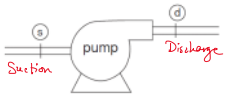
↓ pressure; Temp constant, no phase changes/chemical reaction, negligible heat transfer

Mechanical energy Balance (MEB)

$$\frac{\Delta P}{\rho} + \frac{\Delta\langle v\rangle^2}{2\alpha} + g\Delta z + F = -\frac{W_{s,by}}{m}$$

- ρ: density
- $\langle v\rangle$: average velocity
- $\alpha = 1$ (turbulent flow)
- F: friction term $[\Delta W - Q]$

Pump can add energy via:
+ Increase velocity of fluid
+ Create pressure differential
+ Pump can change fluid height (raise its potential energy)



η = Pump efficiency = $\frac{\text{(useful energy)}}{\text{(total energy)}}$ (energy converted to work (rather than friction))

$= \frac{\left(-\frac{W_{s,by}}{m} - F\right)}{}$

$\left(\frac{-W_{s,bg}}{m}\right)$

Pump brake horsepower: $bhp = -W_{s,bg}$ (Total work delivered by pump)

↓ Dividing by g

$$\therefore \quad \frac{\Delta p}{\rho g} + \frac{\Delta \langle v \rangle^2}{2\alpha g} + \Delta z = \eta \left(\frac{-W_{s,bg}}{mg}\right) = \eta \cdot (bhp)$$

$$\boxed{\left(\frac{P_d}{\rho g} + \frac{\langle v \rangle_d^2}{2g} + z_d\right) - \left(\frac{P_s}{\rho g} + \frac{\langle v \rangle_s^2}{2g} + z_s\right) = \eta \left(\frac{-W_{s,bg}}{mg}\right)}$$

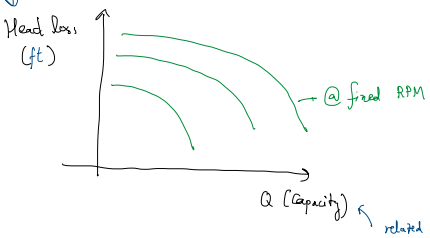
* This form is useful when we know efficiency of pump already (since that takes into account friction loses)

Discharge head (H_d) [=] length

Suction head (H_s) [=] length

$$\underbrace{(H_d - H_s)}_{\text{Head loss}} = \eta \left(\frac{-W_{s,bg}}{mg}\right) = \eta \left[\frac{-bhp}{mg}\right)$$

gives info about η & bhp



Pumping Ability of a CP:
+ Shaft rotational speed (RPM)
+ Shape of vanes
+ flow path

Carrying out the lab:

Characterize relationships b/w discharge flow rate, net head, bhp, efficiency in a CP

↳ Vary Q (via rotameter)
↳ Measure pressure loss → calculate head loss (H_{loss} or $H_{net} = H_d - H_s$)
↳ Non-dimensionalize variables
& then plot them!

Lab Calculations, Discussions & Variables

9.358

$$H_{net} = \text{Head loss} = \frac{P_{out} - P_{in}}{\rho g} + \frac{\langle v \rangle_{out}^2 - \langle v \rangle_{in}^2}{2g\alpha} + \Delta z + \frac{\Sigma}{g}; \qquad F_{piping} = \sum_{j,\text{straight pipe}} \left[4 f_j \frac{L_j}{D_j} \frac{\langle v \rangle_j^2}{2}\right] + \sum_{i,\text{fittings}} \left[n_i k_{f,i} \frac{\langle v \rangle_i^2}{2}\right]$$

where,
f is friction factor given for steady turbulent flow in smooth pipes [9.361]

Non-dimensional variables:

$$C_H = \frac{g H_{net}}{n^2 D^2}$$

n = rotational speed (on pump label)
D = impeller diameter (handout)

$$C_Q = \frac{Q}{nD^3}$$

$$C_P = \frac{bhp}{\rho n^3 D^5}$$

$$\eta = \frac{C_H C_Q}{C_P}$$

+ Plot the characteristic pump curves: (C_H vs. C_Q, C_P vs. C_Q, η vs. C_Q)
(for water & CMC solution)

* can use multiple graphs or one graph w/ secondary axis ↓

Be creative w/ your data graphs to allow readers to understand

+ Discuss the trends! → Why are they like that?
Does it make sense?
How do these compare w/ real pump curve (reference other sources)

+ Error Analysis!
↳ What is error associated w/ all the variables
H_{net}, C_H, C_P, C_Q, η

↳ Are they high/low?
↳ Can they account for the deviations seen in the trends (if any?)?
↳ What are the sources of error in the experiment

Good luck!